

## Uncertain quantities of interest

- Classification of random quantities
- Statistical notation
- Random variables
- Probability functions

## Fundamental classification

Basic types of **quantities of interest** include

- scalars ...temperature, energy, mass, distance, impedance, susceptibility, etc
- vectors ...displacement, velocity, force, magnetization, etc
- functions ...responses, trajectories, landscapes, Lagrangians, Hamiltonians, etc
- measures ...distributions, etc

When definite values are not enough, a QoI is **uncertain**

Uncertain QoIs are classified as

- parametric: these are formed by **finitely** many real scalars ...ie uncertain scalar and vector quantities
- non-parametric: there are formed by **infinitely** many real scalars ...ie uncertain functions and measures

Parametric uncertain quantities are termed **random variables** and studied in parametric Statistics or Statistics

Non-parametric uncertain quantities are termed **stochastic processes** and studied in non-parametric Statistics

BO relies mainly on **uncertain functions** and secondarily on uncertain scalars

## Statistical notation

A shorthand like this

$$W \sim P_\theta$$

indicates an **uncertain**  $W$  and its distribution  $P_\theta$  which depends upon a **certain** parameter  $\theta$

A **probability distribution** is a shorthand for

- what values our uncertain quantity may attain
- how to compute probabilities associated with these values

A shorthand like this

$$W|\theta \sim P_\theta$$

indicates an **uncertain**  $W$  and its distribution  $P_\theta$  which depends upon a **uncertain** parameter  $\theta$

A distribution **requires** parameters that keep it well-defined

## Example 1

A shorthand like

$$W \sim \text{Poisson}(\mu)$$

stands for

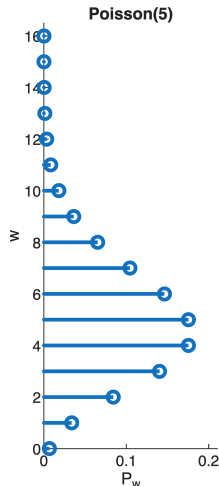
- the **values**  $w$  that  $W$  may attain are integers  $0, 1, 2, \dots$
- the probability **mass** of  $W$  has the form

$$p(w) = \frac{\mu^w}{w!} e^{-\mu}$$

- the **probability** of  $W$  taking any value between some  $w_{\min}$  and  $w_{\max}$  is given by

$$P_{w_{\min}, w_{\max}} = \sum_{w_{\min}}^{w_{\max}} p(w)$$

- the probability mass of  $W$  depends on one **parameter**  $\mu$



## Example 2

A shorthand like

$$W \sim \text{Binomial}(J, \pi)$$

stands for

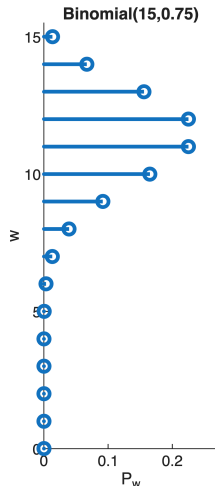
- the **values**  $w$  that  $W$  may attain are integers  $0, 1, \dots, J$
- the probability **mass** of  $W$  has the form

$$p(w) = \binom{J}{w} \pi^w (1 - \pi)^{J-w}$$

- the **probability** of  $W$  taking any value between some  $w_{\min}$  and  $w_{\max}$  is given by

$$P_{w_{\min}, w_{\max}} = \sum_{w_{\min}}^{w_{\max}} p(w)$$

- the probability mass of  $W$  depends on two **parameters**  $J$  and  $\pi$



## Example 3

A shorthand like

$$W \sim \text{Laplace}(m, b)$$

stands for

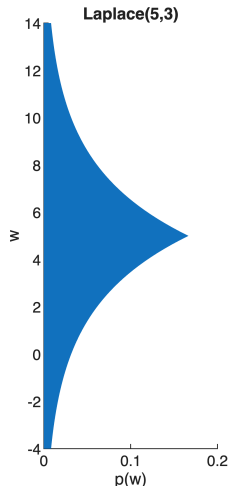
- the **values**  $w$  that  $W$  may attain are reals in  $(-\infty, +\infty)$
- the probability **density** of  $W$  has the form

$$p(w) = \frac{1}{2b} e^{-\frac{|w-m|}{b}}$$

- the **probability** of  $W$  taking any value between some  $w_{\min}$  and  $w_{\max}$  is given by

$$P_{w_{\min}, w_{\max}} = \int_{w_{\min}}^{w_{\max}} p(w) dw$$

- the probability density of  $W$  depends on two **parameters**  $m$  and  $b$



## Example 4

A shorthand like

$$W \sim \text{Uniform}_{[a_{\min}, a_{\max}]}$$

stands for

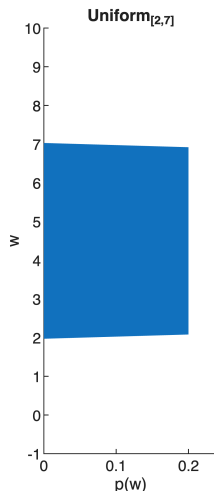
- the **values**  $w$  that  $W$  may attain are real numbers in  $(-\infty, +\infty)$
- the probability **density** of  $W$  has the form

$$p(w) = \begin{cases} 1/(a_{\max} - a_{\min}), & w \in [a_{\min}, a_{\max}] \\ 0, & \text{elsewise} \end{cases}$$

- the **probability** of  $W$  taking any value between some  $w_{\min}$  and  $w_{\max}$  is given by

$$P_{w_{\min}, w_{\max}} = \int_{w_{\min}}^{w_{\max}} p(w) dw$$

- the probability density of  $W$  depends on two **parameters**  $a_{\min}$  and  $a_{\max}$



## Probability distributions

Shorthands like these

$$W \sim P_\theta$$

$$W|\theta \sim P_\theta$$

indicate uncertain quantities. On the **LHS** we have the quantities and on the **RHS** we have their distributions

Distributions are often described **generatively**

$$W \sim \text{Binomial}(J, \pi)$$

Generate  $u_j \sim \text{Uniform}_{[0,1]}$  for  $j = 1, \dots, J$   
Set  $W = \sum_j \mathbb{I}_{u_j < \pi}$

$$W \sim \text{Laplace}(m, b)$$

Generate  $u \sim \text{Uniform}_{[0,1]}$   
Set  $v = u - \frac{1}{2}$   
Set  $W = m + b \log(1 - 2|v|) \text{sgn}(v)$

### Alternative scheme

Generate  $u \sim \text{Uniform}_{[0,1]}$  and  $v \sim \text{Uniform}_{[0,1]}$   
Set  $W = \begin{cases} m + b \log u & v < 1/2 \\ m - b \log u & \text{elsewise} \end{cases}$



## Probability rules

All distributions are probability measures

...we need measure theory

Only **parametric** distributions are characterized by **probability functions**

...we need calculus

The rules to compute parametric probabilities depend on the values of the random variable

- A random variable may be **discrete** and we compute probabilities using a *probability mass function*
- A random variable may be **continuous** and we compute probabilities using a *probability density function*

Probability mass functions  $p(w)$  of **discrete** random variables  $W \sim P$  are non-negative and normalized

$$p(w) \geq 0$$

$$\sum_{\text{all values}} p(w) = 1$$

Probability density functions  $p(w)$  of **continuous** random variables  $W \sim P$  are non-negative and normalized

$$p(w) \geq 0$$

$$\int_{\text{all values}} p(w) dw = 1$$

## Joint probability distributions

Separate equations indicate **independence**

$$\begin{aligned}q &\sim Q \\ r &\sim R\end{aligned}$$

between the specified quantities  **$q \perp r$**

Such a model has a **joint** probability function

$$p(q, r) = p(q)p(r)$$

Conditional equations indicate **dependence**

$$\begin{aligned}q|r &\sim Q_r \\ r &\sim R\end{aligned}$$

between the specified quantities  **$q \not\perp r$**

Such a model has a **joint** probability function

$$p(q, r) = p(q|r)p(r)$$

- No matter what the model is, joint and marginals are related by

$$p(q) = \begin{cases} \int p(q, r) dr, & \text{for continuous } r \\ \sum p(q, r), & \text{for discrete } r \end{cases}$$

$$p(r) = \begin{cases} \int p(q, r) dq, & \text{for continuous } q \\ \sum p(q, r), & \text{for discrete } q \end{cases}$$

- No matter what the model is, joint and conditionals are related by

$$p(q, r) = p(q|r)p(r)$$

$$p(q, r) = p(r|q)p(q)$$